Han Zhang

Assistant Research Scientist

University of Michigan

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Research Interests

Attention; Cognitive Control; Human-AI Interaction

Employment

Assistant Research Scientist *06/2025-present*

University of Michigan Transportation Research Institute

Postdoctoral Research Fellow *06/2020-06/2025*

Department of Psychology, University of Michigan

Education

Doctor of Philosophy, Education and Psychology *09/2014-05/2020*

University of Michigan

- Dissertation: Mind-wandering: What Can We Learn from Eye Movements?

Master of Science, Psychology *09/2014-04/2018*

University of Michigan

Bachelor of Science, Psychology *09/2010-07/2014*

Beijing Normal University

Awards and Grants

As Assistant Research Scientist

Harnessing Driving Behavior to Inform Cognitive Change in the Aging Population *2027-2028*

Building Better Futures, University of Michigan *$50,000*

- Role: PI

Understanding Driver Transitions Across Automated Driving Levels *2026-2027*

General Motors *$219,357*

- Role: PI

Understanding Takeover Performance in Urgent Situations during Level-3 Driving *2025-2027*

UMTRI & Mcity Integration Fund, University of Michigan *$99,971*

- Role: PI

Demonstration and Evaluation of Employer-Based Driving Safety Programs *2024-2029*

National Highway Traffic Safety Administration (NHTSA) *$2,400,000*

- Role: co-I

As Postdoc

Probing Attentional Allocation with a Novel Forced-Response Method *2023*

Standard Grant, National Science Foundation *$659,877*

- Role: Key Personnel (Postdoc)

Investigating Interference-Control in ADHD Using a Novel Forced-Response Method *2023*

R21, National Institute of Mental Health *$429,000*

- Role: Key Personnel (Postdoc)

As Graduate Student

Stanley E. and Ruth B. Dimond Best Dissertation Award *2020*

School of Education, University of Michigan *$500*

- Dissertation: Mind-wandering: What Can We Learn from Eye Movements?

MPA Graduate Student Paper Award *2020*

Midwestern Psychological Association Annual Meeting *$300*

- Scan-paths of mind-wandering during real-world scene perception

ProQuest Distinguished Dissertation Award (Nomination) *2020*

School of Education, University of Michigan

Rackham One-Term Dissertation Fellowship *2019*

Rackham Graduate School, University of Michigan *$17417*

Rackham Graduate Student Research Grant *2019*

Rackham Graduate School, University of Michigan *$3000*

Rackham Conference Travel Grant *2019, 2018, 2017*

Rackham Graduate School, University of Michigan *$1050*

Rackham Summer Training Award *2018*

Rackham Graduate School, University of Michigan *$3700*

Professor & Mrs. Cho-Yee To Travel Grant *2017, 2015*

School of Education, University of Michigan *$500*

As Undergraduate Student

China National Scholarship *2014*

Ministry of Education of the People's Republic of China *$1500*

The First Rank Academic Scholarship *2014, 2013*

Faculty of Psychology, Beijing Normal University *$300*

Beijing Student Research and Entrepreneurial Action Plan *2013*

Beijing Municipal Commission of Education *$1500*

Undergraduate Student Research Grant *2012*

Faculty of Psychology, Beijing Normal University *$280*

Preprints/Under Review

Underlined names indicate undergraduate research mentees.

7. **Zhang, H.**, (2026). *Emergent goal-directed attention in large vision-language models*. arXiv. https://arxiv.org/abs/2609.05517

6. **Zhang, H.**, & Cao, Z. (2026). *From visual search to movement control: A priority field for artificial agents*. arXiv. https://doi.org/10.48550/arXiv.2609.30704

5. **Zhang, H.**, Done, B., Schneider, S., Rajan, N., & Wang, M. (2026). *Who can re-enter the loop? A scoping review on cognitive abilities and task re-engagement in human-AI interaction*. PsyArXiv. https://doi.org/10.31234/osf.io/mhvts_v2

4. Osborne, J., Sellers, J., **Zhang, H.**, Wang, S., Liberatore, S., Sripada, C., Shah, P., & Jonides, J. (2026). *Regardless of Task Demands, Delayed Goal-Directed Processing Persists in Adult ADHD*. PsyArXiv. https://doi.org/10.31234/osf.io/jw2tx_v1

3. Sellers, J., **Zhang, H.**, Jonides, J., & Lee, T. (2026). *Causal evidence for the role of dorsolateral prefrontal cortex in the implementation of cognitive control adjustments*. bioRxiv. https://doi.org/10.64898/2026.09.26.754661

2. Sellers, J., **Zhang, H.**, Zhou, W.-J., Jonides, J., & Lee, T. (2026). *Modeling the temporal dynamics of cognitive control*. PsyArXiv. https://osf.io/preprints/psyarxiv/2gxk4_v1/

1. **Zhang, H.**, Wang, M., Eby, D. W., & St. Louis, R. (2025). *Sensing health and well-being from naturalistic driving behavior*. PsyArXiv. https://doi.org/10.31234/osf.io/jkhfb_v1

Journal Articles

* indicates shared first authorship. Underlined names indicate undergraduate research mentees.

23. Sellers*, J., **Zhang*, H.**, Jonides, J., & Lee, T. G. (2026). Does global slowing explain age effects in inhibitory control? *Psychological Science*, 9567976261479660. https://doi.org/10.1177/09567976261479660

22. York, A. K., Sellers, J., **Zhang, H.**, Vesia, M., Lee, T., & Jonides, J. (2026). Physiological evidence for the time-dependent resolution of response conflicts. *Journal of Cognitive Neuroscience*, *38*(9), 1735–1747. https://doi.org/10.1162/JOCN.a.2616

21. **Zhang, H.**, & Miller, K. F. (2026). I'm trying to read here! How does irrelevant speech affect how you read? *Cognitive Processing*. https://doi.org/10.1007/s10339-026-01346-4

20. Molloy, M. F., Lee, T. G., Jonides, J., **Zhang, H.**, Sellers, J., Heathcote, A., Sripada, C., & Weigard, A. S. (2026). Joint cognitive models reveal sources of robust individual differences in conflict processing. *Computational Brain & Behavior*. https://doi.org/10.1007/s42113-026-00263-1

19. Osborne, J. B., Sellers, J., **Zhang, H.**, Wang, S., Liberatore, S., Sripada, C., Shah, P., & Jonides, J. (2026). Delayed goal-directed processing underlies inhibitory control challenges in adult ADHD. *Scientific Reports*, *16*(1), 13706. https://doi.org/10.1038/s41598-026-42307-3

18. **Zhang, H.**, & Jonides, J. (2026). PupEyes: An interactive Python library for eye movement data processing. *Behavior Research Methods*, *58*(1), 29. https://doi.org/10.3758/s13428-025-02830-z

17. **Zhang, H.**, Brar, E., York, A. K., & Jonides, J. (2025). Are abrupt onsets highly salient? *Journal of Experimental Psychology: Human Perception and Performance*, *51*(7), 911–926. https://doi.org/10.1037/xhp0001329

16. **Zhang, H.**, Sellers, J., Lee, T. G., & Jonides, J. (2025). The temporal dynamics of visual attention. *Journal of Experimental Psychology: General*, *154*(2), 435–456. https://doi.org/10.1037/xge0001661

15. **Zhang, H.**, York, A. K., & Jonides, J. (2025). Attentional capture by abrupt onsets: Foundations and emerging issues. *Journal of Experimental Psychology: Human Perception and Performance*, *51*(3), 283–299. https://doi.org/10.1037/xhp0001275

14. Lee, T. G., Sellers, J., Jonides, J., & **Zhang, H.**, (2025). The forced-response method: A new chronometric approach to measure conflict processing. *Behavior Research Methods*, *57*(1), 15. https://doi.org/10.3758/s13428-024-02516-y

13. Fansher, M., Lalwani, P., Adkins, T. J., **Zhang, H.**, Quirk, M., Carlson, M., Boduroglu, A., Lewis, R. L., Jonides, J., & Shah, P. (2025). A brief intervention to improve reasoning about accumulation. *Journal of Experimental Psychology: Applied*, *31*(2), 99–125. https://doi.org/10.1037/xap0000532

12. **Zhang, H.**, Miller, K. F., & Jonides, J. (2024). How does mind-wandering affect distractor suppression? *Visual Cognition*, *32*(9-10), 1027–1044. https://doi.org/10.1080/13506285.2024.2347605

11. Adkins, T. J., **Zhang, H.**, & Lee, T. G. (2024). People are more error-prone after committing an error. *Nature Communications*, *15*(1), 1–11. https://doi.org/10.1038/s41467-024-50547-y

10. **Zhang, H.**, Abagis, T. R., Steeby, C. J., & Jonides, J. (2024). Lingering on distraction: Examining distractor rejection in adults with ADHD. *Visual Cognition*, *32*(9-10), 910–924. https://doi.org/10.1080/13506285.2024.2328379

9. **Zhang, H.**, Miyake, A., Osborne, J., Shah, P., & Jonides, J. (2023). A d factor? Understanding trait distractibility and its relationships with ADHD symptomatology and hyperfocus. *PLOS ONE*, *18*(10), e0292215. https://doi.org/10.1371/journal.pone.0292215

8. Osborne, J. B., **Zhang, H.**, Carlson, M., Shah, P., & Jonides, J. (2023). The association between different sources of distraction and symptoms of attention deficit hyperactivity disorder. *Frontiers in Psychiatry*, *14*. https://doi.org/10.3389/fpsyt.2023.1173989

7. Fansher, M., Adkins, T. J., Lalwani, P., Boduroglu, A., Carlson, M., Quirk, M., Lewis, R. L., Shah, P., **Zhang, H.**, & Jonides, J. (2022). Icon arrays reduce concern over COVID-19 vaccine side effects: A randomized control study. *Cognitive Research: Principles and Implications*, *7*(1), 38. https://doi.org/10.1186/s41235-022-00387-5

6. **Zhang, H.**, Anderson, N. C., & Miller, K. F. (2022). Scene meaningfulness guides eye movements even dur-

ing mind-wandering. *Attention, Perception, & Psychophysics*, *84*(4), 1130–1150. https://doi.org/10.3758/s13414-021-02370-6

5. **Zhang, H.**, Abagis, T. R., & Jonides, J. (2021). The malleability of attentional capture. *Visual Cognition*, *29*(9), 571–574. https://doi.org/10.1080/13506285.2021.1915903

4. **Zhang, H.**, Anderson, N. C., & Miller, K. F. (2021). Refixation patterns of mind-wandering during real-world scene perception. *Journal of Experimental Psychology: Human Perception and Performance*, *47*(1), 36–52. https://doi.org/10.1037/xhp0000877

3. **Zhang, H.**, Miller, K. F., Sun, X., & Cortina, K. S. (2020). Wandering eyes: Eye movements during mind wandering in video lectures. *Applied Cognitive Psychology*, *34*(2), 449–464. https://doi.org/10.1002/acp.3632

2. **Zhang, H.**, Qu, C., Miller, K. F., & Cortina, K. S. (2020). Missing the joke: Reduced rereading of garden-path jokes during mind-wandering. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, *46*(4), 638–648. https://doi.org/10.1037/xlm0000745

1. **Zhang, H.**, Miller, K., Cleveland, R., & Cortina, K. (2018). How listening to music affects reading: Evidence from eye tracking. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, *44*(11), 1778–1791. https://doi.org/10.1037/xlm0000544

Book Chapters and Conference Proceedings

Underlined names indicate undergraduate research mentees.

4. Brar, E., **Zhang, H.**, & Jonides, J. (2025). Overt attentional suppression of highly salient onsets. *Vision Sciences Society Annual Meeting Abstract*, *25*, 2731. https://doi.org/10.1167/jov.25.9.2731

3. **Zhang, H.**, York, A. K., & Jonides, J. (2024). Less salient, more capture: The curious case of abrupt onsets. *Vision Sciences Society Annual Meeting Abstract*. Vision Sciences Society Annual Meeting. https://doi.org/10.1167/jov.24.10.1296

2. Michal, A., Fansher, M., Sun, X., **Zhang, H.**, & Shah, P. (2022). Cognitive development: Applications to education. In A. O'Donnell, N. C. Barnes, & J. Reeve (Eds.), *The oxford handbook of educational psychology* (1st ed., pp. 31–72). Oxford University Press. https://doi.org/10.1093/oxfordhb/9780199841332.013.7

1. **Zhang, H.**, & Jonides, J. (2021). Pre-trial gaze stability predicts momentary slips of attention. *ACM CHI '21 Workshop on Eye Movements as an Interface to Cognitive State*. https://psyarxiv.com/bv2uc/

Selected Presentations

Underlined names indicate undergraduate research mentees.

25. Done, B., Schneider, S., Rajan, N., & **Zhang, H.**, (2026, August). *Who can re-enter the loop? A scoping review*. [2nd-Place Winner] 2026 UMTRI Student Poster Competition. Ann Arbor, MI, USA.

24. Rajan, N., Schneider, S., Done, B., & **Zhang, H.**, (2026, August). *Can machine learning predict L3 driving takeover performance from gaze and cognitive traits?* [1st-Place Winner] 2026 UMTRI Student Poster Competition. Ann Arbor, MI, USA.

23. Schneider, S., Done, B., Rajan, N., & **Zhang, H.**, (2026, August). *Who can reengage better during level-3 automated driving?* [Poster] 2026 SURE Symposium. Ann Arbor, MI, USA.

22. Brar, E., **Zhang, H.**, & Jonides, J. (2024, November). *Does a temporally predictable distractor facilitate visual search?* [Poster]. The Psychonomic Society's 65th Annual Meeting. New York City, NY, USA.

21. **Zhang, H.**, Weissman, D. H., & Jonides, J. (2024, November). *What's 'reactive' about proactive suppression?* [Talk]. The Psychonomic Society's 65th Annual Meeting. New York City, NY, USA.

20. **Zhang, H.**, York, A. K., & Jonides, J. (2024, May). *Less salient, more capture: The curious case of abrupt onsets*. [Talk]. VSS 2024. St. Pete Beach, FL, USA.

19. Zhou, W.-J., York, A. K., **Zhang, H.**, & Jonides, J. (2024, May). *Assessing ADHD effects on working memory manipulation: Evidence from drift diffusion modeling*. [Poster] Object Perception, Attention, and Memory (OPAM) Conference. New York City, NY, USA.

18. Brar, E., Jonides, J., & **Zhang, H.**, (2024, April). *Rejecting a temporally predictable salient distractor*. [Poster]. Undergraduate Research Opportunity Program (UROP) Symposium. Ann Arbor, MI, USA.

17. **Zhang, H.**, Sellers, J., Lee, T. G., & Jonides, J. (2023, November). *Modeling the temporal dynamics of visual attention*. [Talk]. The Psychonomic Society's 64th Annual Meeting. San Francisco, CA, USA.

16. **Zhang, H.**, & Jonides, J. (2023, July). *Pretrial stability predicts momentary lapses of attention*. [Talk]. Current Issues in Mind-Wandering Research Conference. Heidelberg, Germany.

15. **Zhang, H.**, Adkins, T., Lee, T., & Jonides, J. (2022, November). *How does the priority map change over time?* [Talk]. The Psychonomic Society's 63nd Annual Meeting. Boston, MA, USA.

14. **Zhang, H.**, Miyake, A., Shah, P., & Jonides, J. (2021, November). *Uncovering the structure of individual differences in distractibility*. [Talk]. The Psychonomic Society's 62nd Annual Meeting. Virtual Conference.

13. **Zhang, H.**, Anderson, N. C., & Miller, K. F. (2020, November). *Mind-wandering during scene perception: On the role of meaning and salience*. [Poster]. The Psychonomic Society's 61st Annual Meeting. Virtual Conference.

12. Hwang, Y., & **Zhang, H.**, (2020, May). *Predicting attention during mind-wandering using meaning and saliency maps*. [Poster]. The Association for Psychological Science Annual Meeting. Chicago, IL, USA.

11. **Zhang, H.**, Anderson, N. C., & Miller, K. F. (2020, April). *Scan-paths of mind-wandering during real-world scene perception*. [Graduate Student Research Award]. The Annual Meeting of the Midwestern Psychological Association (Conference Canceled Due to Covid-19).

10. **Zhang, H.**, Qu, C., Miller, K. F., & Cortina, K. S. (2019, November). *Reduced re-reading of garden-path jokes during mindless reading*. [Poster]. The Psychonomic Society's 60th Annual Meeting. Montreal, Québec, Canada.

9. **Zhang, H.**, Miller, K. F., Cortina, K. S., & Jiang, T. (2019, June). *Mind-wandering in college classrooms: A mobile eye-tracking study*. [Poster]. SARMAC XIII. Cape Cod, MA, USA.

8. Fischer, A., & **Zhang, H.**, (2019, May). *Examining the effect of word predictability during mindless reading.* [Poster]. The Association for Psychological Science Annual Meeting. Washington DC, USA.

7. **Zhang, H.**, & Shah, P. (2019, May). *What can iPhone's screen time tell about your cognitive functioning?* [Poster]. The Association for Psychological Science Annual Meeting. Washington, DC, USA.

6. **Zhang, H.**, Miller, K. F., & Sun, X. (2019, March). *Scan-paths of mind-wandering during video lectures*. [Poster]. The International Convention of Psychological Science. Paris, France.

5. **Zhang, H.**, & Miller, K. F. (2018, November). *How irrelevant speech affects reading: The role of word predictability*. [Poster]. The Psychonomic Society's 59th Annual Meeting. New Orleans, LA, USA.

4. Qu, C., & **Zhang, H.**, (2018, May). *All joking aside: Mind-wandering impairs processing of garden-path jokes.* [Poster]. The Association for Psychological Science Annual Meeting. New Orleans, LA, USA.

3. **Zhang, H.**, Miller, K. F., & Sun, X. (2018, May). *The wandering eyes: Mind-wandering during video lectures is associated with oculomotor behaviors*. [Poster]. The Association for Psychological Science Annual Meeting. San Francisco, CA, USA.

2. Fraija, N., Evans, M., **Zhang, H.**, & Miller, K. (2018, April). *Tracking attention in the classroom: An mobile eye-tracking study*. [Poster]. Undergraduate Research Opportunity Program (UROP) Symposium. Ann Arbor, MI, USA.

1. **Zhang, H.**, Miller, K. F., Cleveland, R., & Cortina, K. S. (2016, May). *How listening to music affects reading: Evidence from eye tracking*. [Poster]. The Association for Psychological Science Annual Meeting. Chicago, IL, USA.

Invited Talks

2. **Zhang, H.**, (2026, May). *Measurement of drivers' re-engagement readiness*. Mcity Annual Research Review. University of Michigan.

1. **Zhang, H.**, (2026, April). *What can we learn from eye movements?* Experimental Curiosity Lecture Series. Lawrence Technological University.

Teaching and Mentoring Experience

Primary Instructor

EDUC 391: Educational Psychology and Human Development *09/2017 - 12/2017*

Participated in course design; taught weekly lectures; graded assignments and exams; held office hours

Graduate Student Instructor

PSYCH 111: Introduction to Psychology *01/2019 - 04/2019*

Taught three one-hour classes per week; graded student essays and exams; held office hours

PSYCH 240: Introduction to Cognitive Psychology *09/2016 - 12/2016*

Taught three one-hour classes per week; graded student essays and exams; held office hours

PSYCH 457: Research in Educational and Cross-Cultural Settings *2016 - 2020*

Graded student essays and exams; held office hours; supervised student research projects

Global Course Connection: Beijing Normal University *2015 - 2019*

Supervised joint research projects between US and Chinese students; organized cultural exchange activities

Undergraduate Research Opportunity Program (UROP) Mentor

Mentored students: Madelyn Evans (2018), Nicole Fraija (2018), Yun-Gi Hwang (2019), Alexander Fischer (2019), Esha Brar (2023) *2018 - 2025*

Mentored undergraduate students conducting research projects.

Scholarship, Transferable skills, Academia & Research (STAR) Scholars Mentor

Mentored students: Brianna Westpfahl (2019), Michael Cloud (2019), Julliana Ramirez-Matias (2019), Rachel Kushner (2020), Ruby Zheng (2020), Skylar Brodnan (2020) *2018 - 2020*

Mentored underrepresented students on academic research engagement and transferable skills. More info: https://lsa.umich.edu/psych/undergraduates/star-scholars-program.html

Summer Undergraduate Research in Engineering (SURE) Mentor

Mentored students: Stefanie Schneider (2026) *2026*

Mentored undergraduate students working full-time on research projects in human-AI interaction.

Honors Thesis Mentor

Mentored students: Bianca Done *2026-2027*

Open-Source Software

- PupEyes: An Interactive Python Library for Eye Movement Data Processing

Grant-Writing Experience

Understanding Driver Transitions Across Automated Driving Levels *2026*

General Motors *Funded*

- Role: PI

Establishing the Mechanistic Role and Translational Potential of Task-Positive Brain Network Flexibility as a Transdiagnostic Neural Underpinning of Cognitive Control *2026*

R01, National Institute of Mental Health *Pending*

- Role: co-I (PI: Weigard)

Gaze-Aware Support for Effective Human-Autonomy Teaming *2026*

Automotive Research Center *Pending*

- Role: PI

Characterization of Attentive, Unimpaired, Non-distracted, Competent Driver *2026*

NHTSA, USDOT *Pending*

- Role: PI

DMS-based Countermeasures to Mitigate Distraction Risks *2026*

NHTSA, USDOT *Pending*

- Role: co-I

Harnessing Driving Behavior to Inform Cognitive Change in the Aging Population *2026*

University of Michigan *Funded*

- Role: PI

Understanding Takeover Performance in Urgent Situations during Level-3 Driving *2025*

UMTRI & Mcity Integration Fund, University of Michigan *Funded*

- Role: PI

Conference: Object Perception, Attention, and Memory Meeting *2024*

National Science Foundation *Funded*

- Role: Conceptualization; original draft preparation; reviewing & editing (PI: Hout)

Investigating Interference-Control in ADHD Using a Novel Forced-Response Method *2023*

R21, National Institute of Mental Health *Funded*

- Role: Key Personnel (PI: Jonides)

Probing Attentional Allocation with a Novel Forced-Response Method *2023*

National Science Foundation *Funded*

- Role: Key Personnel (PI: Jonides)

Professional Service

Object Perception, Attention, and Memory (OPAM) Conference Organizer

- OPAM 31, November 16th 2023, San Francisco, CA, US
- OPAM 32, November 21th 2024, New York City, NY, YS
- `https://opamconference.com/`

2026 Automated Transportation Symposium Session Organizer

- SAE Level 3 Features: Coming Soon to a Vehicle Owned by You

Ad Hoc Reviewer

- Memory & Cognition
- NeuroImage
- Visual Cognition
- Cognitive Science
- Scientific Studies of Reading
- Frontiers in Psychiatry
- Psychological Research
- Cognition
- Scientific Reports
- Journal of Experimental Psychology: Human Perception and Performance
- PLOS One
- Journal of Memory and Language
- Neurobiology of Aging
- Behavior Research Methods
- Psychonomic Bulletin & Review
- Attention, Perception, & Psychophysics
- Traffic Psychology and Behaviour
- IEEE Intelligent Transportation Systems Transaction

- Autism
- Journal of Gerontology: Psychological Sciences

References

John Jonides, Ph.D. *jjonides@umich.edu*

Edward E. Smith Professor of Psychology and Neuroscience; Co-Director, functional MRI Center, University of Michigan

Taraz G. Lee, Ph.D. *tarazlee@umich.edu*

Assistant Professor of Psychology, University of Michigan

Priti Shah, Ph.D. *priti@umich.edu*

Department Chair, Arthur Thurnau Professor of Psychology, Education, and Information, University of Michigan

Kevin F. Miller, Ph.D. *kevinmil@umich.edu*

Professor of Psychology and Education, University of Michigan